

Test Plan — Arteri E-Learning (Stuudi)

Project: Arteri E-Learning Platform (frontend repo: `stuudi-frontend`) **Repo:** <https://github.com/irzadzulhika29/stuudi-frontend> **Document Version:** v1.0 **Date:** 2026-06-21 **Prepared By:** Muhammad Irza Dzulhika (QA Intern Candidate) **Template:** Based on BINUS University Test Plan Framework (Mulyanto, 2018)

Document Information

Field	Value
Project Name	Arteri E-Learning Platform (codename: Stuudi)
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Prepared By	Muhammad Irza Dzulhika
Reviewed By	(to be assigned)
Approved By	(to be assigned)
Frontend Repo	github.com/irzadzulhika29/stuudi-frontend
Backend Repo	github.com/azmiagr/lms-project (out of scope)
Live URL	https://stuudi.vercel.app

1. Introduction (Pendahuluan)

1.1. Rencana Pengujian

Dokumen ini menjabarkan rencana pengujian komprehensif untuk platform **Arteri E-Learning** — aplikasi web berbasis Next.js 16 yang dirancang untuk mempersiapkan siswa menghadapi olimpiade sains nasional. Pengujian akan mencakup 4 area utama: **CBT (Computer-Based Testing) engine**, **autentikasi & role management**, **manajemen kursus & konten**, dan **dashboard admin/teacher**. Pengujian dilakukan pada frontend (Next.js) dengan fokus pada functional, integration, dan end-to-end testing.

1.2. Tujuan Pengujian

Tujuan pengujian platform Arteri adalah: - **Memastikan CBT engine** (fitur paling kritis) berjalan sempurna untuk ujian olimpiade nasional - Tab switch detection + 3-lives anti-cheat system berfungsi - Auto-save jawaban saat network failure - Timer countdown akurat - 5 tipe soal (single, multiple, true/false, matching, short_answer) berjalan sesuai spek - **Memvalidasi role-based access** (student vs teacher vs admin) sesuai `src/middleware.ts` - **Memverifikasi integrasi API** dengan backend `lms-project` (60+ endpoint di `API_ENDPOINTS`) - **Mengurangi regression** dengan E2E automation + unit test yang sudah ada - **Memenuhi standar kualitas** untuk apply magang QA IT

1.3. Metodologi Pengujian

Metode	Tools	Coverage
Black-box functional testing	Manual + Postman (API)	Semua fitur user-facing
Component unit testing	Vitest + Testing Library	13 test files existing (CBT-focused)
End-to-end testing	Playwright	5 spec files existing
Security testing	OWASP ZAP, manual probing	Auth flow, role bypass, JWT
Performance testing	k6 (planned)	CBT engine load test
Accessibility testing	axe-core (planned)	Form labels, keyboard nav, contrast

1.4. Objek Pengujian

Layer	Technology	Test Scope
Frontend	Next.js 16.0.10 (App Router) + React 19.2.1	Full UI + client logic
API Client	Axios 1.13 (with JWT interceptor)	All 60+ endpoints
State Management	Redux Toolkit + React Query	Auth slice, exam slice, UI slice
CBT Engine	Custom React + Redux (5 question types)	Full functional
Auth Middleware	Next.js middleware.ts (JWT decode)	Role-based redirect
Backend (proxy)	lms-project (Go/Python)	E2E only (out of scope)

1.5. Arsitektur Sistem yang Diuji



1.6. Pembagian Sistem untuk Pengujian Unit dan Integrasi

Test Level	Target	Tools	Lokasi
Unit test	Pure functions, hooks, isolated components	Vitest + Testing Library	src/**/__tests__/
Component integration	Component + Redux + hooks	Vitest + Testing Library	src/**/__tests__/
API integration	Axios calls mocked via MSW or mock	Vitest	src/services/__tests__/
E2E	Full flow (UI → API → DB)	Playwright	e2e/*.spec.ts
Manual	Visual regression, exploratory	Manual + screenshots	Staging environment

2. Test Items (Item Pengujian)

2.1. Item yang Akan Diuji

A. Authentication & Authorization (HIGH PRIORITY)

ID	Item	Tipe	Modul Path
AUTH-01	Login form validation & submit	UI/Functional	<code>src/features/auth/login/</code>
AUTH-02	JWT token storage (cookie + localStorage)	Integration	<code>src/middleware.ts</code>
AUTH-03	Role-based redirect (teacher → admin, student → dashboard)	Integration	<code>src/middleware.ts</code>
AUTH-04	401 auto-logout via Axios interceptor	Integration	<code>src/shared/api/api.ts</code>
AUTH-05	Cross-role access block (student → admin routes)	Security	<code>src/middleware.ts</code>

B. CBT (Computer-Based Testing) Engine (HIGHEST PRIORITY — CRITICAL)

ID	Item	Type	Modul Path
CBT-01	Exam intro screen & start action	UI/Functional	<code>src/features/user/cbt/</code>
CBT-02	Question rendering (5 types)	UI/Functional	<code>src/features/user/cbt/components/</code>
CBT-03	Answer state management (Redux)	State	<code>src/features/user/cbt/utls/answerState.ts</code>
CBT-04	Auto-save on answer change	Integration	<code>src/features/user/cbt/hooks/useAutoSave.ts</code>
CBT-05	Timer countdown + auto-submit	Critical	<code>src/features/user/cbt/hooks/useExamTimer.ts</code>
CBT-06	Tab switch detection + 3-lives system	Critical	<code>src/features/user/cbt/hooks/useFullscreenGuard.ts</code> + <code>examService.recordTabSwitch</code>
CBT-07	Question navigation (skip, mark, review)	UI/Functional	<code>src/features/user/cbt/components/QuestionNavigation.tsx</code>
CBT-08	Exam summary screen	UI/Functional	<code>src/features/user/cbt/components/ExamSummary.tsx</code>
CBT-09	Submit exam & result	Integration	<code>examService.submitExam</code>
CBT-10	Resume from snapshot (localStorage cache)	Edge case	<code>examService.transformExamToReduxPayload</code>
CBT-11	Math rendering (KaTeX)	UI	<code>src/shared/components/math/</code>
CBT-12	Image question rendering	UI	<code>QuestionCard.tsx</code>

C. Student Dashboard & Course Management (MEDIUM PRIORITY)

ID	Item	Type	Modul Path
STU-01	Student dashboard (upcoming exam, courses)	UI/Functional	<code>src/features/user/dashboard/</code>
STU-02	Browse courses (search, filter, pagination)	UI/Functional	<code>src/features/user/courses/</code>
STU-03	Course detail & topic tree	UI/Functional	<code>src/features/user/courses/</code>
STU-04	Course content (text/media/quiz blocks)	UI/Functional	<code>src/features/user/courses/</code>
STU-05	Quiz flow (start, answer, submit, result)	Integration	<code>COURSES.QUIZ.*</code> endpoints
STU-06	Notes per topic (CRUD)	UI/Functional	<code>COURSES.TOPIC_NOTES</code> endpoint
STU-07	Notifications (list, mark read, delete)	UI/Functional	<code>COURSES.NOTIFICATIONS*</code>
STU-08	Team details view	UI/Functional	<code>src/features/user/teams/</code>

D. Admin/Teacher Dashboard (MEDIUM PRIORITY)

ID	Item	Tipe	Modul Path
ADM-01	Admin dashboard home (stats)	UI/Functional	<code>src/features/admin/dashboard/home/</code>
ADM-02	Courses management (CRUD)	Integration	<code>TEACHER.COURSES*</code> endpoints
ADM-03	Topic & content management	Integration	<code>TEACHER.ADD_TOPIC/UPDATE_TOPIC/DELETE_TOPIC</code>
ADM-04	Quiz authoring (Tiptap editor)	UI/Functional	<code>src/features/admin/dashboard/courses/</code>
ADM-05	Block editor (text/media/quiz)	UI/Functional	<code>TEACHER.ADD_*_BLOCK</code> endpoints
ADM-06	Exam management (CRUD)	Integration	<code>TEACHER.EXAM*</code> endpoints
ADM-07	Exam dashboard (analytics)	UI/Functional	<code>TEACHER.GET_EXAM_DASHBOARD</code>
ADM-08	Cheating report (per exam)	UI/Functional	<code>src/app/(admin)/dashboard-admin/cheating-report/</code>
ADM-09	Disqualified participants list	UI/Functional	<code>TEACHER.DISQUALIFIED_PARTICIPANTS</code>
ADM-10	Participants management (bulk upload)	Integration	<code>TEACHER.ADD_PARTICIPANT_BULK</code>

E. Public / Landing (LOW PRIORITY)

ID	Item	Tipe	Modul Path
PUB-01	Landing page (home)	UI/Visual	<code>src/features/public/landing/</code>
PUB-02	Product page	UI/Visual	<code>src/app/(landing)/produk/</code>
PUB-03	About us page	UI/Visual	<code>src/app/(landing)/tentang-kami/</code>

2.2. Item yang TIDAK Akan Diuji

- **Backend API implementation** (lms-project) — di luar scope frontend testing
- **Email delivery** (notification system) — out of scope
- **Payment gateway** — tidak ada di project ini
- **Mobile native app** — tidak ada, hanya web responsive
- **Performance under 10K concurrent users** — akan diukur terpisah dengan k6
- **Penetration testing** — out of scope, akan dilakukan oleh security team
- **Cross-browser legacy** (IE11, Safari < 14) — target browser modern saja

2.3. Definisi Istilah Penting

Istilah	Definisi
CBT	Computer-Based Testing — sistem ujian digital olimpiade
JWT	JSON Web Token untuk authentication
Attempt	Sesi pengerjaan ujian oleh satu siswa untuk satu exam
Lives	"Nyawa" siswa (max 3) — berkurang saat terdeteksi tab switch
Snapshot	Cache soal di localStorage untuk recovery saat resume
Block	Unit konten dalam course (text / media / quiz)
Topic	Sub-bab dalam Course
Course	Mata pelajaran / paket pembelajaran
Examination	Ujian yang dijadwalkan teacher untuk student
Disqualified	Student yang kehabisan lives (3 tab switches)

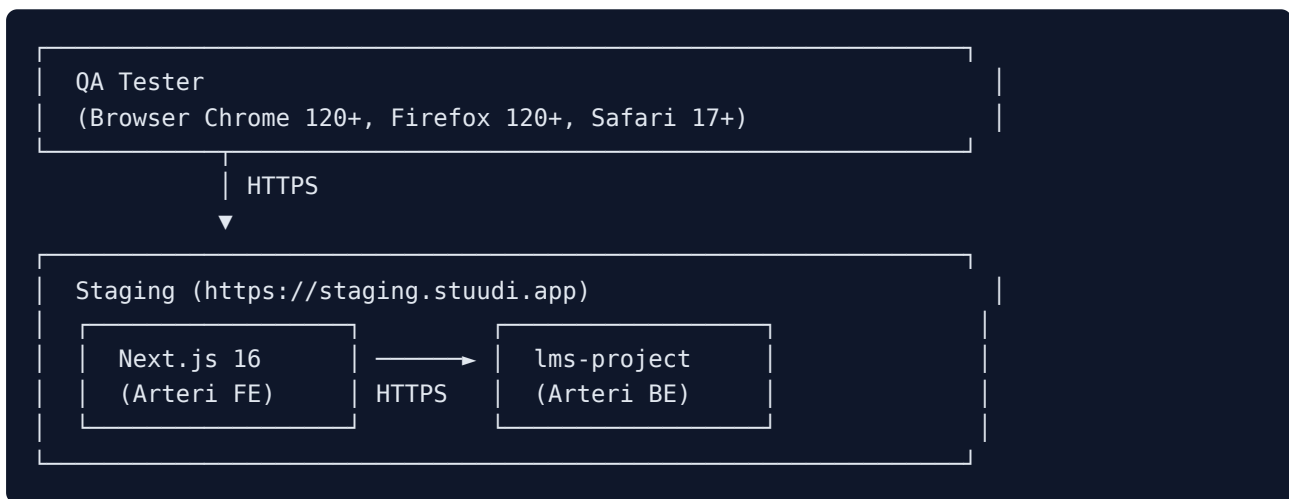
3. Special Terminology (Istilah Khusus)

Istilah	Definisi
Tab switch	Aksi keluar dari halaman ujian ke tab lain (window.blur)
Fullscreen guard	Hook yang memaksa fullscreen + detect escape
Auto-save	Mekanisme simpan jawaban otomatis setiap X detik
Resume	Lanjut ujian setelah disconnect (data dari localStorage)
Question type	Tipe soal: <code>single</code> , <code>multiple</code> , <code>true_false</code> , <code>matching</code> , <code>short_answer</code>
Matching pair	Pasangan sisi kiri-kanan untuk soal matching
Side	Posisi opsi di soal matching (<code>left</code> / <code>right</code>)
Bearer token	Format <code>Authorization: Bearer <token></code>
Route group	Next.js folder <code>(name)</code> yang tidak masuk URL (untuk layout grouping)
App Router	Next.js 13+ routing system (vs legacy Pages Router)

4. Testing Location (Lokasi Pengujian)

Environment	URL	Tujuan	Akses
Local	<code>http://localhost:3000</code> + <code>http://localhost:8080/api/v1</code>	Development testing	Developer laptop
Staging	<code>https://staging.stuudi.app</code> (planned)	Pre-production QA	QA team
Production	<code>https://stuudi.vercel.app</code>	Smoke test only	Public
Mock backend	MSW (Mock Service Worker)	Unit test isolation	Test runner

Diagram setup pengujian:



5. Testing Schedule (Jadwal Pengujian)

No	Kegiatan	Tanggal Mulai	Tanggal Selesai	PIC	Status
1	Test Plan Review	2026-06-21	2026-06-22	Irza	✅ Done
2	Test Case Creation (existing)	2026-06-19	2026-06-20	Irza	✅ Done (76 TC)
3	API Test Case Creation (existing)	2026-06-20	2026-06-20	Irza	✅ Done (100 TC)
4	E2E Automation Setup (Playwright)	2026-06-20	2026-06-21	Irza	✅ Done (5 specs)
5	Iterasi 1: CBT engine full test	2026-06-22	2026-06-28	QA Team	🕒 Planned
6	Iterasi 2: Auth & role-based	2026-06-29	2026-07-03	QA Team	🕒 Planned
7	Iterasi 3: Course & content	2026-07-06	2026-07-10	QA Team	🕒 Planned
8	Iterasi 4: Admin dashboard	2026-07-13	2026-07-17	QA Team	🕒 Planned
9	Regression Test (full)	2026-07-20	2026-07-22	QA Team	🕒 Planned
10	UAT (User Acceptance Test)	2026-07-23	2026-07-24	Stakeholder	🕒 Planned
11	Sign-off & Release v1.0	2026-07-25	2026-07-25	PM	🕒 Planned

6. Testing Quality Criteria (Kriteria Kualitas Pengujian)

Kriteria	Target
Kesiapan environment (staging)	100% deployed & smoke test passed
Ketersediaan test data	Akun student + teacher test tersedia di staging
Build stability	Build sukses tanpa critical blocker
Kesiapan test case	176 TC finalized (76 UI + 100 API) ✅
Code coverage (unit)	Target minimal 70% untuk CBT engine
E2E coverage	Minimal 1 happy-path + 1 sad-path per fitur critical
Dokumentasi	Test plan + test summary report

7. Entry Criteria (Kriteria Masuk)

- [x] Spesifikasi kebutuhan (requirements) sudah didokumentasikan di **API_ENDPOINTS**
- [x] Test case UI sudah dibuat (76 TC) di Google Sheet
- [x] Test case API sudah dibuat (100 TC) di Google Sheet
- [x] Environment pengujian (local + staging) tersedia

- [x] Test data (akun student + teacher test) sudah dibuat
- [x] Tools testing terinstall (Vitest, Playwright, Postman, VS Code)
- [x] Backend `lms-project` running di `localhost:8080` (untuk E2E)
- [x] E2E automation setup (5 spec files) sudah dibuat

8. Exit / Suspension Criteria (Kriteria Keluar / Penangguhan)

Suspension (pengujian ditunda): - Build gagal deploy ke staging - Backend `lms-project` down > 30 menit
 - Ada critical bug yang block > 50% test case CBT - Test environment tidak stabil (sering timeout)

Resume (pengujian dilanjutkan): - Build sudah stabil di staging - Backend sudah running normal - Critical bug sudah diperbaiki - Environment sudah stabil minimal 1 jam

9. Completion Criteria (Kriteria Penyelesaian)

- [] **100% CBT test case** dieksekusi (highest priority, 18 TC)
- [] Minimal **95% test case** total dieksekusi (167 dari 176)
- [] **0 critical bug** open
- [] **0 high-priority bug** open (atau di-defer untuk v1.1)
- [] Semua **regression test** passed
- [] Test summary report sudah dibuat
- [] Sign-off dari stakeholder

10. Environmental Requirements (Kebutuhan Lingkungan)

10.1. Perangkat Keras (Hardware)

Device	Spec	Jumlah
Laptop Tester	RAM 16GB, SSD 512GB, i5 ke atas	1 (QA)
Smartphone (Android)	Android 13+, 6GB RAM (responsive test)	1
Tablet (iPad)	iPad Air, iOS 17+ (responsive test)	1

10.2. Perangkat Lunak (Software)

Software	Versi	Kegunaan
Node.js	18+	Runtime
Google Chrome	120+	Browser testing (primary)
Firefox	120+	Cross-browser
Safari	17+	macOS testing
Playwright	1.61+	E2E test runner
Vitest	4.0+	Unit test runner
Postman	Latest	API testing (manual)
Newman	Latest	Postman CLI (automation)
VS Code	Latest	Code editor
k6	Latest	Load testing (planned)
axe-core	Latest	Accessibility testing (planned)

10.3. Jaringan

- Internet min. 10 Mbps (untuk akses staging & API eksternal)
- Tidak ada VPN yang dibutuhkan (semua public)

10.4. Konfigurasi Environment

Variable	Value	Keterangan
<code>BASE_URL</code>	<code>http://localhost:3000</code>	Frontend dev
<code>NEXT_PUBLIC_BASE_API</code>	<code>http://localhost:8080/api/v1</code>	Backend API
<code>API_BASE_URL</code>	<code>http://localhost:8080/api/v1</code>	E2E test config
<code>STUDENT_USER</code>	<code>student.test@stuudi.id</code>	Test student
<code>TEACHER_USER</code>	<code>teacher.test@stuudi.id</code>	Test teacher
Cookie name	<code>access_token</code>	JWT storage

11. Test Systems (Sistem Pengujian)

Komponen	Deskripsi	Status
Test case UI	Google Sheet "Test Cases" — 76 TC (47 positif, 29 negatif)	✓ Ada
Test case API	Google Sheet "API Test Cases" — 100 TC (64 positif, 36 negatif)	✓ Ada
Test plan	Dokumen ini (<code>test-plan.md</code>)	✓ Ada
E2E scripts	<code>e2e/*.spec.ts</code> — 5 spec files (49 test)	✓ Ada
Unit tests	<code>src/**/*.__tests__/*.test.ts(x)</code> — 13 files (35 test)	✓ Ada
Test runner	Vitest config + Playwright config	✓ Ada
CI/CD pipeline	<code>.github/workflows/e2e.yml</code> (Playwright)	✓ Ada
API client (Postman)	<i>Belum dibuat</i> — planned	⌚ Planned
k6 load test	<i>Belum dibuat</i> — planned	⌚ Planned

12. External Factors (Faktor Eksternal)

- **Backend** `lms-project` — harus running untuk E2E test
- Disimpan terpisah (bukan bagian dari frontend repo)
- Endpoint contract harus stabil; breaking change perlu dikoordinasikan
- **Supabase** — digunakan untuk image storage (CDN `dl1vucwgezsuwktkwxd.supabase.co`)
- Reliability: tergantung Supabase SLA
- **Image CDN (unsplash, ui-avatars)** — untuk avatar & demo content
- Reliability: 99.9% SLA
- **Vercel** — hosting frontend
- Reliability: 99.99% SLA
- **Tiptap editor** — untuk authoring konten (dari npm)
- Update berkala, perlu cek breaking changes

13. Team Composition (Komposisi Tim)

Nama	Peran	Tanggung Jawab
Irza (Muhammad Irza Dzulhika)	QA Lead / Tester	Test planning, execution, automation
Alfi (dev)	Frontend Developer	Bug fixing, E2E spec review
Dzaki	Support contact	Issue escalation
Nopal	Support contact	Issue escalation
Pak Aminul (PKL supervisor)	Reviewer	Test plan review (academic)

External Participants: Guru/dosen pembimbing olimpiade (untuk UAT CBT)

Hand-off Points: - Developer → QA: setiap sprint / iterasi (via PR) - QA → Stakeholder: setiap release untuk UAT

14. Revision Handover Management (Manajemen Penyerahan Revisi)

- **Version control:** Git (GitHub)
- **Build deployment:** otomatis via Vercel (per push ke `main`)
- **QA testing window:** minimal 1x24 jam setelah build baru di staging
- **Changelog format:** ``` [v0.1.X] 2026-06-21
- **FIXED:** Bug #123 - Login error 500 on expired token
- **ADDED:** CBT tab switch counter UI
- **CHANGED:** Redesign QuestionCard with new layout
- **TESTED:** 18 CBT test cases executed ```
- **Notifikasi:** GitHub Issues + Slack `#qa-coordination` (planned)

15. Test Suite Execution (Eksekusi Test Suite)

Strategi per fase:

Fase	Test Suite	Strategi	Tools
Smoke	5 critical-path tests	Automated on every PR	Playwright
CBT regression	18 CBT test cases (UI) + 7 CBT E2E	Automated on every PR to CBT module	Vitest + Playwright
Auth regression	10 auth test cases (UI) + 10 auth E2E	Automated	Playwright
Functional	Per fitur (per modul)	Manual + Postman untuk API	Manual + Postman
Regression	Full suite	Automated nightly	Playwright + Vitest
Performance	Load test (k6)	Manual weekly	k6 (planned)
Security	OWASP Top 10 checklist	Manual + automated	OWASP ZAP (planned)
Accessibility	axe-core scan	Automated	axe-core (planned)

Cara eksekusi (Arteri-specific):

```
# Setup lokal
cd stuudi-frontend
cp .env.example .env.local
# Update NEXT_PUBLIC_BASE_API di .env.local

# Install dependencies
npm install

# Install Playwright browsers
npx playwright install

# Unit tests
npm test                # Run once
npm run test:watch      # Watch mode
npm run test:coverage   # With coverage report

# E2E tests (per module)
npm run test:e2e:auth   # Login & auth
npm run test:e2e:cbt    # CBT (priority)
npm run test:e2e:api    # API integration
npm run test:e2e:admin  # Admin dashboard
npm run test:e2e:student-dashboard # Student area

# E2E cross-browser
npm run test:e2e:chromium
npm run test:e2e:firefox

# E2E with UI for debugging
npm run test:e2e:ui
npm run test:e2e:debug
npm run test:e2e:headed

# Full E2E suite
npm run test:e2e

# API testing (after Postman collection ready)
newman run arteri-api.postman_collection.json
```


16. Potential Risks (Risiko Potensial)

Risiko	Dampak	Probabilitas	Mitigasi
Backend lms-project down	E2E test gagal semua	Medium	Mock backend dengan MSW untuk unit test; fallback ke manual
Staging environment down	Testing delay	Medium	Local Docker compose backup (planned)
CBT tab switch logic berubah	Existing test jadi invalid	Low	Baca PR diff sebelum re-run
JWT secret rotation	Test token invalid	Low	Auto-generate test token via login API
Kurangnya test data untuk matching question	Coverage rendah	Medium	Auto-generate matching question di seed script
Tight deadline	Test terburu-buru	High	Prioritas CBT (highest impact)
Tim QA single person (Irza)	Coverage tidak maksimal	High	Risk-based testing; fokus CBT
Backend API contract berubah tanpa notice	Integration test gagal	Medium	Strict TS types → tangkap saat compile
Math rendering (KaTeX) edge cases	Soal matematika broken	Low	Dedicated test untuk math
Concurrent tab switch (race condition)	Lives counter bug	Medium	Stress test dengan 5 tab

17. Documentation of Changes (Dokumentasi Perubahan)

Versi	Tanggal	Penulis	Perubahan
v1.0	2026-06-21	Irza	Initial Test Plan — berdasarkan analisis codebase arteri_elearning

18. Appendix (Lampiran)

Dokumen pendukung: - **Test Case UI (76 TC)** — [Google Sheet](#) - **Test Case API (100 TC)** — [Google Sheet](#) - **Frontend repo** — <https://github.com/irzadzulhika29/stuudi-frontend> - **Backend repo** — <https://github.com/azmiagr/lms-project> - **E2E spec files** — `e2e/*.spec.ts` di repo - **Unit test files** — `src/**/__tests__/_` di repo - **API contract** — `src/shared/config/constants.ts` (`API_ENDPOINTS`)

19. FAQ (Pertanyaan Umum)

Q: Mengapa CBT engine diprioritaskan? A: CBT adalah fitur paling kritis — ini untuk **ujian olimpiade nasional**. Bug di CBT bisa menggagalkan ujian ribuan siswa. Lives system + auto-save + timer = critical path.

Q: Bagaimana cara testing tab switch detection di E2E? A: Pakai Playwright `page.evaluate()` untuk trigger `window.dispatchEvent(new Event('blur'))` atau switch ke tab lain via `context.newPage()`. Verify lives counter decrement via API call.

Q: Kapan pakai unit test vs E2E test? A: **Unit test** untuk logic (timer, answer state, normalization). **E2E test** untuk full flow (start exam → answer → save → submit → result). Jangan duplikat.

Q: Bagaimana cara test auto-save tanpa tunggu 30 detik? A: Mock timer di Vitest (`vi.useFakeTimers()`). Di E2E, set interval auto-save pendek (override via env) atau tunggu real time.

Q: Apa beda `useExamTimer` vs `useAutoSave` vs `useFullscreenGuard`? A: - `useExamTimer` — countdown ujian, auto-submit saat habis - `useAutoSave` — simpan jawaban ke server periodik - `useFullscreenGuard` — paksa fullscreen + detect escape/tab switch

Q: Kenapa pakai Vitest bukan Jest? A: Vitest lebih cepat (Vite-based), zero-config untuk Vite/Next.js, dan support TypeScript native tanpa babel.

Q: Test mana yang harus automated duluan? A: Prioritas (sesuai risiko): 1. CBT tab switch + auto-save + submit (critical path) 2. Auth login + role-based redirect 3. Course enrollment + content access 4. Admin exam creation

Q: Berapa lama test plan ini berlaku? A: Berlaku per release (v1.0). Update setiap ada perubahan arsitektur besar (misal: migrasi Next.js Pages Router → App Router sudah dilakukan; migrasi berikut mungkin ke React Server Component penuh).

Referensi

- Mulyanto, A. (2018). *Test Plan*. BINUS University – School of Information Systems.
- Myers, G. J., Sandler, C., & Badgett, T. (2012). *The Art of Software Testing* (3rd ed.). John Wiley & Sons, Inc.
- IEEE 829 — Standard for Software Test Documentation
- ISTQB Foundation Level Syllabus v4.0
- Next.js Testing Best Practices — <https://nextjs.org/docs/app/building-your-application/testing>
- Playwright Documentation — <https://playwright.dev/docs/intro>
- Vitest Documentation — <https://vitest.dev/>

Change Log (Dokumen Ini)

Versi	Tanggal	Perubahan
v1.0	2026-06-21	Initial Test Plan — instantiated dari template BINUS, diisi spesifik untuk Arteri E-Learning berdasarkan analisis codebase